

SUMMER INTERNSHIP REPORT PRESENTATION

Front-End Web Development at Tripekart

Building scalable travel e-commerce components, interactive itineraries, dynamic filtering, and high-performance UI layers.

SUBMITTED BY

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PROGRAM & SEMESTER

B.Tech CSE / IT

5th Semester / 3rd Year

ORGANIZATION

Tripekart (tripekart.com)

May 2026 – September 2026

Company Profile — Tripekart

Domain: Travel & Tourism Tech



Specialized Tourism Portal

Online travel platform offering group backpacking trips, weekend getaways, and customized itineraries across Spiti Valley, Ladakh, Himachal, and Meghalaya.

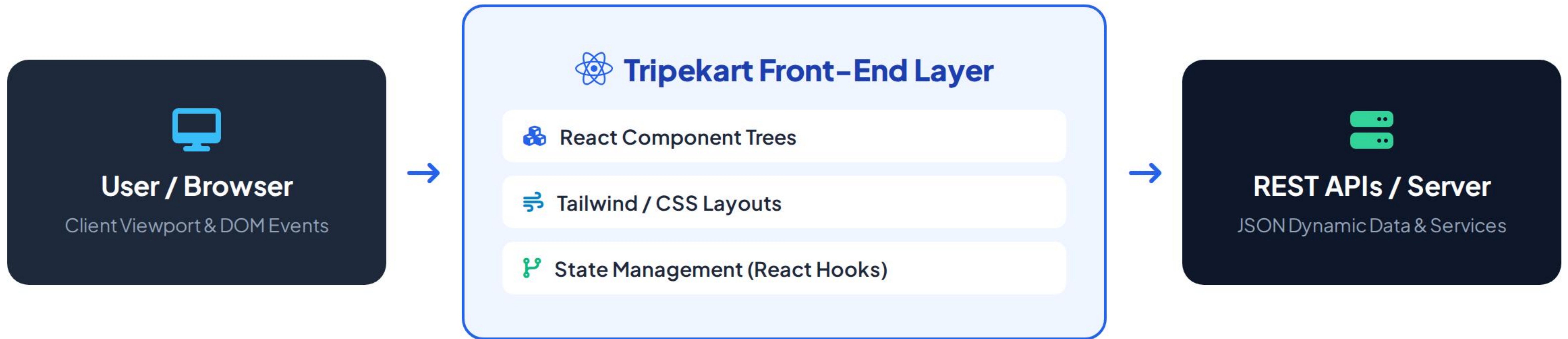
Core Platform Offerings

Delivers transparent pricing models, day-wise schedules, dynamic package filter engines, and seamless lead booking inquiries.

Technical Focus Areas

Responsive UI/UX design, interactive component trees, high-performance media delivery, and lead capture system integrations.

Tripekart Web System Architecture



Internship Objectives



Responsive Interface Design

Build intuitive UI components for package listings, detail views, and lead generation forms.



Cross-Device Polish

Ensure fluid mobile-first layouts using modern flexbox, CSS grids, and breakpoints.



REST API Integration

Connect dynamic views with RESTful endpoints for live pricing and booking workflows.



Code Efficiency

Minimize re-renders, optimize visual media, and maintain modular React components.

Primary Driving Goal: Direct improvement in user engagement and booking conversion rates.

Tools & Tech Stack Used

Core Web Tech

HTML5 & CSS3: Semantic structural markup & flexbox layouts

JavaScript (ES6+): Async fetch calls, DOM logic & dynamic validation

Frameworks & Styling

React.js: Reusable component trees, routing & state management

Tailwind CSS: Utility-first styling & unified theme design

Tools & Workflow

Git & GitHub: Feature branching & pull request code reviews

Postman & DevTools: REST API payload testing & profile debugging



Modern industry-standard stack optimized for high-speed travel e-commerce platforms.

Month-Wise Work Summary (May – Sept 2026)

MAY 2026

Onboarding & UI Setup

- Analyzed legacy codebase
- Built buttons & navbar
- Standardized breakpoints

JUNE 2026

Modules & Validation

- Redesigned lead forms
- Regex phone/email checks
- Accordion itinerary views

JULY 2026

API & Dynamic Filters

- Connected live REST endpoints
- Budget & duration filters
- Modal state management

AUG – SEPT 2026

Optimization & QA

- Implemented native lazy loading
- Cross-browser testing
- Developer documentation

Structured sprint progression focused on incremental UI stability and component reuse.

Interactive Itinerary Accordion Component

ItineraryAccordion.jsx (React + Tailwind)

```
const ItineraryAccordion = ({ dayNumber, title, description, highlights }) => {
  const [isOpen, setIsOpen] = useState(false);
  return (
    <div className="border-b border-slate-200 my-2">
      <button onClick={() => setIsOpen(!isOpen)} className="w-full flex justify-between">
        <span>Day {dayNumber}: {title}</span>
        <span>{isOpen ? '-' : '+'}</span>
      </button>
      {isOpen && (
        <div className="p-4 bg-white text-slate-600">
          <p>{description}</p>
          <div>
            {highlights}
          </div>
        </div>
      )}
    </div>
  );
};
```

Conditional Rendering

Utilizes state-driven conditional DOM node allocation to reduce memory usage during long tour browsing.

Zero Page Re-renders

Expands and collapses day-wise schedules instantly without full page reloads or layout jumping.

Mobile Performance

Maintains fluid 60fps scrolling on low-end mobile hardware during complex itinerary views.

Deployed across primary tour package pages including Spiti Valley Backpacking and Ladakh Expeditions.

Key Engineering Challenges & Solutions



1. Image-Heavy Load Speed

CHALLENGE

High-resolution gallery images caused slow page loads on mobile networks.

SOLUTION IMPLEMENTED

Added native lazy loading (`loading="lazy"`), WebP conversion, and fixed aspect ratios to prevent CLS.



2. Form State & Validation

CHALLENGE

Multi-step lead forms allowed invalid contact inputs, degrading lead quality.

SOLUTION IMPLEMENTED

Wrote client-side Regex checks for phone/email and locked submit buttons until constraints passed.

Directly boosted lead intake accuracy while decreasing bounce rates on mobile landing pages.

Key Learnings & Takeaways



Production Code Standards

Gained hands-on practice structuring component trees, modular UI logic, and scalable folder organization.



UI/UX Empathy

Understood how minor UI tweaks—like instant button feedback and dynamic hints—drive customer conversions.



Agile Team Workflow

Participated in daily standups, Git pull request code reviews, and cross-functional sprint planning.

Successfully bridged classroom CSE concepts at IILM University with real-world industry execution.

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Thank You!

Questions & Discussion

Student: Akshay Yadav (Roll No: 2410030258)

Guide: Faculty Guide & Head of Department (CSE)

References: Tripekart (tripekart.com) | MDN Web Docs | React.js Docs



Department of Computer Science & Engineering, IILM University

Image Sources



<https://www.tripgurugo.com/wp-content/uploads/2026/06/SpitiValleyinJulyATravelGuideForTheBestExperience.webp>

Source: www.tripgurugo.com